

TINGRI, China

WMO: 556640

Lat: 28.6542N

Lon: 87.1202E

Elev: 4300

StdP: 59.27

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.5	(c) -15.9	(d) -35.6	(e) 0.2	(f) -3.8	(g) -33.7	(h) 0.3	(i) -4.4	(j) 15.6	(k) 1.9	(l) 13.9	(m) 1.9	(n) 0.3	(o) 40	(p) 0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.3	(c) 20.7	(d) 7.6	(e) 19.7	(f) 7.6	(g) 18.7	(h) 7.5	(i) 9.9	(j) 16.9	(k) 9.4	(l) 16.4	(m) 9.0	(n) 15.9	(o) 3.6	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	11.0	12.0	6.9	10.6	11.6	6.4	10.3	11.4	42.7	17.0	41.2	16.5	39.8	16.1	15.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	10.0	8.5	-21.2	23.0	2.0	1.0	-22.6	23.7	-23.8	24.3	-24.9	24.8	-26.3	25.5
			-22.3	11.0	1.6	1.0	-23.5	11.7	-24.4	12.3	-25.3	12.9	-26.4	13.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-6.0	-4.4	-1.0	2.9	7.7	12.4	13.0	12.2	10.5	4.3	-1.0	-4.3
	DBStd	7.27	2.98	2.81	2.56	2.42	2.47	1.79	1.66	1.58	1.67	3.07	2.28	3.08
	HDD10.0	2501	496	403	341	212	79	3	1	3	14	176	329	445
	HDD18.3	5270	754	636	599	462	331	179	166	192	236	434	579	703
	CDD10.0	273	0	0	0	0	7	74	94	69	28	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	2.4	2.6	3.2	3.4	3.3	2.9	2.3	1.8	1.6	1.7	1.9	1.9	2.0
Precipitation	PrecAvg	346	2	2	2	5	14	36	120	123	35	5	1	1
	PrecMax	469	7	4	5	15	35	94	243	210	103	19	17	6
	PrecMin	206	0	0	0	1	2	9	60	53	9	1	0	0
	PrecStd	74	2	1	1	3	9	21	46	42	24	4	3	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	10.8	14.1	16.2	20.3	22.2	22.2	20.2	19.4	16.7	12.5	11.0
		MCWB	-1.5	-0.9	1.0	2.1	5.2	7.5	8.3	8.7	7.1	4.2	0.2	-1.0
	2%	DB	6.9	8.0	11.1	14.2	18.4	20.9	20.7	19.0	18.0	15.2	10.7	8.6
		MCWB	-3.0	-2.3	-0.6	1.5	4.9	7.3	8.4	8.6	6.6	3.7	-0.6	-2.0
	5%	DB	4.7	5.9	9.0	12.5	16.7	19.8	19.4	18.0	17.0	13.8	9.2	6.7
		MCWB	-4.2	-3.4	-1.6	0.8	4.1	7.3	8.2	8.5	6.6	2.9	-1.2	-2.8
	10%	DB	2.6	4.2	7.3	10.8	15.2	18.7	18.1	16.9	16.0	12.3	7.5	4.7
		MCWB	-5.2	-4.3	-2.3	0.2	3.3	7.1	8.3	8.4	6.5	2.1	-2.0	-3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	-0.4	1.5	4.1	7.9	10.2	10.4	10.9	9.7	6.9	2.3	0.4
		MCDB	8.5	9.2	11.4	12.7	16.2	18.9	17.8	17.5	16.3	13.5	9.6	8.4
	2%	WB	-2.6	-2.0	0.1	2.6	6.6	9.4	9.8	9.9	8.6	5.2	0.4	-1.3
		MCDB	6.1	7.0	9.6	11.6	15.3	17.9	16.7	16.3	15.2	13.0	8.7	7.3
	5%	WB	-3.9	-3.2	-1.0	1.6	5.4	8.8	9.4	9.4	8.0	4.1	-0.7	-2.6
		MCDB	4.2	5.4	8.1	10.8	14.2	17.1	16.3	15.7	14.5	11.7	7.8	6.2
	10%	WB	-5.1	-4.2	-2.1	0.7	4.4	8.2	9.0	8.9	7.4	3.1	-1.7	-3.8
		MCDB	2.3	3.8	6.6	9.8	13.1	16.4	16.0	15.2	13.9	10.5	7.0	4.5
Mean Daily Temperature Range	5% DB	MDBR	17.8	17.0	16.3	15.6	14.7	13.3	10.3	9.7	11.9	15.9	18.3	19.1
		MCDBR	20.8	20.0	19.0	17.9	16.6	14.9	12.5	11.4	13.3	16.6	20.4	21.4
		MCWBR	13.4	12.3	10.5	8.4	6.5	5.0	3.6	3.6	4.3	6.8	11.3	13.2
	5% WB	MCDBR	20.1	18.8	17.9	16.1	14.4	11.7	9.9	9.5	10.7	14.4	19.1	21.1
		MCWBR	13.3	12.0	10.6	8.5	6.6	4.5	3.8	3.7	4.3	6.4	11.3	13.6
Clear-Sky Solar Irradiance	taub	0.173	0.190	0.230	0.280	0.283	0.283	0.286	0.280	0.258	0.209	0.177	0.165	
	taud	2.711	2.622	2.501	2.376	2.439	2.506	2.544	2.566	2.592	2.704	2.761	2.802	
	Ebn at Noon	1123	1112	1071	1014	1003	996	992	1002	1023	1076	1109	1125	
	Edh at Noon	71	85	104	123	116	108	103	100	94	78	67	62	
All-Sky Solar Radiation	RadAvg	4.16	4.84	5.82	6.40	6.75	6.49	5.62	5.33	5.53	5.57	4.71	4.18	
	RadStd	0.32	0.44	0.30	0.47	0.43	0.44	0.36	0.25	0.31	0.31	0.24	0.25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.34	N/A	N/A	N/A	N/A	N/A	N/A	-180	+51	N/A
Regional (1 neighbor)	+0.34	N/A	N/A	N/A	N/A	N/A	N/A	-162	+73	N/A

Nomenclature: See separate page